



ARTIFICIAL INTELLIGENCE

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1. Executive Summary:

Deepfakes—*AI-synthesized videos and images that fabricate a person's likeness*—represent one of the most critical threats to digital trust today. They are used to spread disinformation, commit fraud, and undermine reputations. This project, DeepFake Detector, proposes and implements a multi-algorithm AI system capable of detecting deepfake media in images and videos. The system utilizes a CNN for visual artifacts, a novel Emotion-Consistency Forensics (ECF) module for detecting audio-visual mismatches, and Grad-CAM for explainability, all deployed through an interactive *Streamlit* web application.

2. Problem Statement:

The proliferation of deepfake technology poses serious challenges including political manipulation, identity theft, and non-consensual imagery. *Existing deepfake detectors focus primarily on visual artifacts*. However, modern GAN/diffusion-based deepfakes have reduced pixel-level artifacts significantly. A multi-modal, behavior-aware detection approach is now essential to identify sophisticated fakes that standard visual-only models miss.

3. Project Objectives:

- Detect deepfake images and videos using AI algorithms.
- Implement an EfficientNet-B0 CNN to detect fake faces in images/videos.
- Introduce an ECF module that cross-checks facial emotion against voice tone.
- Generate Grad-CAM heatmaps showing why the AI flagged media as fake.
- Deploy a simple web interface (*Streamlit*) to upload and test media.

4. System Architecture & Methodology:

The system uses a layered multi-algorithm pipeline:

- **Input Layer:** Image or Video upload via *Streamlit* Web UI.
- **Preprocessing Layer:** Face detection (MTCNN), frame extraction, and audio separation (MoviePy).
- **Visual Analysis:** EfficientNet-B0 CNN detects blur boundaries and skin texture artifacts, outputting a Visual Fake Score (0 to 1).
- **ECF Module ★:** Performs a three-way cross-modal consistency check (FER for facial emotion, Whisper for speech-to-text, VADER for sentiment analysis) to output an ECF Inconsistency Score.
- **Fusion Layer:** A weighted average (60% Visual, 40% ECF) computes the final verdict and confidence percentage.
- **Output / XAI:** Verdict label, confidence score, Grad-CAM saliency map, and ECF text report.

5. AI Algorithms & Implementation:

5.1 EfficientNet-B0 CNN: A pre-trained EfficientNet-B0 convolutional neural network, fine-tuned on the Deepfake and Real Images dataset (Kaggle). It serves as the primary visual classifier.

5.2 Grad-CAM: Generates a heatmap overlay on the original image. Red areas indicate where the AI found the strongest evidence of deepfake artifacts, making the model's decision explainable.

5.3 Emotion-Consistency Forensics (ECF) ★: The novel contribution of this project. Because deepfake pipelines synthesize visual and audio layers separately, they produce subtle but detectable mismatches in emotional congruence. ECF cross-checks the facial emotion against the speech sentiment. If a mismatch is found (e.g., smiling face + negative words), an ECF Inconsistency Score is generated.

6. Dataset:

The publicly available "Deepfake and Real Images" dataset from Kaggle was used for training and evaluation. It contains over 10,000 images split between Real and Fake classes.

7. Technology Stack

- **Code:** Python 3.10
- **AI Framework:** PyTorch
- **CNN Model:** EfficientNet-B0 (via timm)
- **Face Detection:** MTCNN (facenet-pytorch)
- **Facial Emotion:** FER library
- **Speech to text:** OpenAI Whisper (small model)
- **Sentiment Analysis:** VADER (NLTK)
- **Explainability:** Native PyTorch Grad-CAM
- **Interface:** Streamlit

8. Results & Evaluation:

The EfficientNet-B0 model was trained for *3 epochs* on a CPU-based system, achieving a validation accuracy of **92.84%** on the first epoch, stabilizing at over **80% overall accuracy** on unseen test data. The Grad-CAM successfully highlighted manipulated regions in images. The ECF module successfully transcribed audio, detected sentiment, and flagged emotional mismatches in video inputs.

9. Limitations:

- ECF *requires clear audio*; it defaults neutral in silent videos or heavy background noise.
- FER works poorly on extreme angles, occlusions, or low-resolution faces.
- Real emotional acting (e.g., sarcasm) may trigger false positives.

- Real-time performance is not guaranteed on CPU; GPU acceleration is required for faster video processing.

10. Conclusion:

This project successfully demonstrates a multi-algorithm approach to deepfake detection. By combining visual CNN analysis with the novel Emotion-Consistency Forensics module, the system can detect both pixel-level artifacts and behavioral inconsistencies that standard detectors miss. The integration of Grad-CAM ensures the AI's decisions are transparent and explainable.